

The Copying Machine

Bonus · Cell Cycle and Cell Division

Starring Cella the root-tip cell 🌱

A BAIO storybook by Narendra Sir · Biology Lecturer

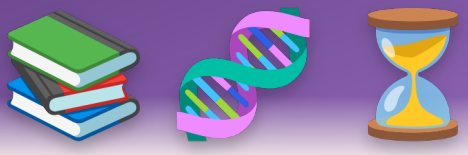


An Urgent Message

Deep in a growing root tip lived a young cell named Cella. One day an urgent message arrived: "Divide carefully. One mistake can change everything." Cella knew she was part of the **cell cycle** — the sequence by which a cell copies its DNA and divides into two. "A human cell takes about 24 hours," she remembered. "Yeast can do it in just 90 minutes!"

💡 **REMEMBER**

The cell cycle has two phases: interphase (preparation) and the M phase (division).



The Busy Rest

First came **interphase** — no rest at all! In the **G1 phase**, Cella grew and made proteins. In the **S phase** she copied her DNA, so the amount doubled from 2C to 4C — but the **chromosome number** stayed the same. In the **G2 phase** she made proteins for division. Some cells stop dividing and enter a quiet **G0 phase**.

💡 **REMEMBER**

Interphase takes more than 95 percent of the cell cycle. In animal cells the centriole also duplicates in the S phase.

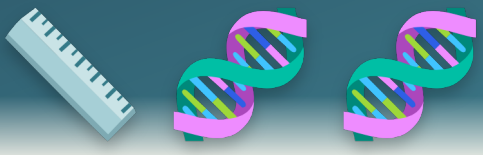


Prophase: Packing Up

The M phase began with **prophase**. Cella's long chromatin threads coiled and condensed into visible **chromosomes**, each made of two **sister chromatids** joined at the **centromere**. The spindle began to form. Slowly the Golgi complex, ER, **nucleolus** and **nuclear envelope** disappeared from view.

💡 **REMEMBER**

Mitosis is called equational division because the chromosome number of the daughter cells equals that of the parent.

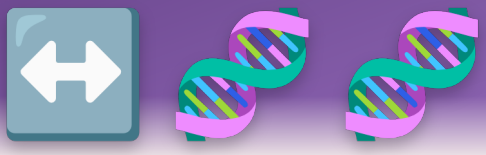


Metaphase: Line Up!

In **metaphase**, **spindle fibres** attached to disc-shaped **kinetochores** on each centromere. The chromosomes moved and lined up neatly along the middle of the cell — the **metaphase plate**. "This is the best stage to study the shape of chromosomes," whispered a microscope from above.

💡 **REMEMBER**

Metaphase is identified by all chromosomes lying at the equator, each attached to spindle fibres from both poles.



Anaphase: Pull Apart

In **anaphase**, each centromere split. The **sister chromatids** — now called daughter chromosomes — were pulled to opposite poles by the shortening spindle fibres, centromere first. "Exactly one copy to each side," counted Cella. "No mistakes allowed!"

💡 **REMEMBER**

At anaphase, centromeres split and chromatids separate; chromatids move to opposite poles.



Telophase and the Split

In **telophase**, the chromosomes reached the poles and uncoiled. Nuclear envelopes and nucleoli formed again — two new nuclei! Then came **cytokinesis**. "Animal cells pinch in with a furrow," said Cella, "but I am a plant cell with a rigid wall." A **cell plate** grew from the centre outward and became the new wall.

💡 **REMEMBER**

Karyokinesis without cytokinesis forms a multinucleate syncytium, e.g. the liquid endosperm of coconut.



When Copying Goes Wrong

Cella remembered her teacher's warning. If DNA is copied with an error, the change is a **mutation**. If cells divide out of control, a **tumour** can form. "That is why there are checkpoints in the cycle," she said. Mitosis is important for **growth**, for **repair** of tissues and for keeping the right **nucleo-cytoplasmic ratio**.

💡 **REMEMBER**

Meristems (apical and lateral) of plants divide continuously by mitosis and cause growth throughout life.



Two Healthy Cells

Cella was now two healthy, identical **daughter cells**, each with the same number of chromosomes. Far away, in a flower's anther, a different division was happening — **meiosis**, which halves the chromosome number to make gametes and mixes genes by **crossing over**. "Both divisions keep life going," Cella smiled, as the root tip grew a little deeper.

💡 **REMEMBER**

Meiosis: one DNA replication and two divisions give four haploid cells. Crossing over occurs at pachytene of prophase I.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

cell cycle p.1 · interphase p.2 · G1 phase p.2 · S phase p.2 · chromosome number p.2 · G2 phase p.2 · G0 phase p.2 · prophase p.3 · chromosomes p.3 · sister chromatids p.3 · centromere p.3 · nucleolus p.3 · nuclear envelope p.3 · metaphase p.4 · spindle fibres p.4 · kinetochores p.4 · metaphase plate p.4 · anaphase p.5 · telophase p.6 · cytokinesis p.6 · cell plate p.6 · mutation p.7 · tumour p.7 · growth p.7 · repair p.7 · nucleo-cytoplasmic ratio p.7 · daughter cells p.8 · meiosis p.8 · crossing over p.8



QUICK QUIZ

Do you remember?

1. During which phase does DNA replicate?

- G1 phase
- S phase
- G2 phase
- M phase

2. What separates during anaphase of mitosis?

- Homologous chromosomes
- Sister chromatids
- Nuclear envelopes
- Cell walls

3. What is the result of mitosis?

- Four haploid cells
- Two genetically identical cells
- One large cell
- Gametes only

4. Chromosomes line up at the equator during:

- Prophase
- Metaphase
- Anaphase
- Telophase

5. In plant cells, cytokinesis takes place by:

- Furrow formation
- Cell plate formation
- Budding
- Fission

Answers: 1-S phase · 2-Sister chromatids · 3-Two genetically identical cells · 4-Metaphase · 5-Cell plate formation